

Positive Recurrence of Weakly Reversible Binary Reaction Networks with Three Species and Two Linkage Classes

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Abstract

We prove a parameter-independent recurrence theorem for stochastic mass-action reaction networks. If every complex has molecularity at most two, there are at most three species, and there are at most two weakly reversible linkage classes, then the minimal continuous-time Markov chain is nonexplosive and every state in every closed irreducible component is positive recurrent, for every positive rate vector. More generally, the same result holds when these bounds are satisfied after exact reduction on the chosen closed class. The proof combines an exact fixed-class projection, a symbolic tier and affine-class reduction, all-clock stopped Foster estimates on the remaining boundary faces, and a finite disjoint support identity. Finite computation is used only for finite support, tier, affine, and set identities; no stochastic estimate is obtained by searching orientations, reaction histories, or population boxes. A separately compiled technical supplement contains the complete byte-frozen local proofs and their reproducibility manifest.

1 Introduction

Weak reversibility is a central structural condition in reaction network theory. Deterministically, it is closely tied to persistence and complex balance. Stochastically, however, recurrence is sensitive to boundary faces, disabled sources, and physical holding times. A descending deterministic tier need not be enabled, and recurrence of an embedded jump chain need not control the continuous-time process. The chemical recurrence conjecture asks whether weak reversibility alone forces positive recurrence on every closed irreducible class of the stochastic mass-action chain.

Several important cases are known. Complex-balanced systems have class-restricted product-form stationary laws [3], and are nonexplosive [2]. Anderson and Kim gave a tier criterion that forces positive recurrence [4, Theorem 9]. For one linkage class, Anderson, Cappelletti, and Kim proved the binary case when every species occurs as a pure unary or binary complex, together with a broader proper-tier criterion [1]. These results do not cover disabled mixed top complexes on boundary faces.

This paper establishes the conjectured conclusion in the following finite but genuinely three-dimensional regime.

Theorem 1.1 (Main theorem). *Let $(\mathcal{S}, \mathcal{C}, \mathcal{R}, \kappa)$ be a finite weakly reversible stochastic mass-action reaction network. Suppose every complex has total molecularity at most two, $|\mathcal{S}| \leq 3$, and the reaction graph has at most two linkage classes. For every positive rate vector κ , the minimal continuous-time Markov chain is nonexplosive, and every state in every closed irreducible population class is positive recurrent.*

The intrinsic version, Theorem 3.3, allows more ambient species or linkages when all but three coordinates are constant and all but two projected linkages are inactive or merge on the fixed closed class.

The proof is deliberately separated into analytic and finite components. The analytic component treats arbitrary strongly connected orientations and arbitrary positive rates. It includes every physical clock, the reaction that causes a cutoff, the actual terminal population, and physical duration. The finite component verifies only an exact disjoint partition of a finite support universe. This distinction is essential: a catalog of reaction words cannot prove the uniform stochastic estimates used here.

The local estimates are substantial, so their complete proofs are supplied in the accompanying Technical Proof Supplement. The main article proves the global reduction, records the exact hypotheses and conclusions of each local theorem, proves the symbolic tier and affine interfaces, and shows how the standalone one- and two-linkage results imply Theorem 1.1.

Literature and priority boundary. We use only the cited published theorems and the proofs contained in the accompanying supplement. We make no priority claim for any overlapping case beyond the explicit mathematical statement proved here.

2 Stochastic reaction networks

Let $\mathcal{S} = \{S_1, \dots, S_d\}$ and identify a complex $y = \sum_i y_i S_i$ with $y \in \mathbb{N}_0^d$. A labelled reaction is $y \rightarrow z$ with rate constant $\kappa_{yz} > 0$. At population $x \in \mathbb{N}_0^d$ its stochastic mass-action propensity is

$$\lambda_{y \rightarrow z}(x) = \kappa_{yz}(x)_y, \quad (x)_y = \prod_{i=1}^d \frac{x_i!}{(x_i - y_i)!}, \quad (1)$$

with the convention that the propensity is zero unless $x \geq y$. Parallel labelled channels are retained. The generator is

$$\mathcal{A}f(x) = \sum_{y \rightarrow z \in \mathcal{R}} \lambda_{y \rightarrow z}(x) \{f(x + z - y) - f(x)\}. \quad (2)$$

The network is *binary* when $|y|_1 \leq 2$ for every complex. Its linkage classes are the connected components of the undirected complex graph. It is weakly reversible when every linkage is strongly connected as a directed graph. A closed irreducible population class Γ is a closed communicating class of the chain.

Positive recurrence is always a classwise assertion here. Finite or absorbing classes are included. For an infinite irreducible class it is equivalent to finite mean positive return to one, and hence every, state.

3 Exact reduction on a closed class

Fix a closed irreducible class Γ . Call coordinate i constant if $x_i = m_i$ for all $x \in \Gamma$. Delete constant coordinates, delete a linkage having no enabled source anywhere on Γ , project all remaining complexes, and merge projected linkages that share a projected vertex. Retain parallel labels and discard projected zero-displacement labels.

Lemma 3.1 (Fixed-class conjugacy). *The reduced stochastic mass-action chain is exactly conjugate to the original chain restricted to Γ . It is weakly reversible, and projection cannot increase the number of dynamic species or active linkage classes.*

Proof. Let L contain a complex y_0 enabled at $x \in \Gamma$, and put $r = x - y_0 \geq 0$. Along a directed path $y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_k$ in L , the state before the j th reaction is $r + y_j$. Thus the whole path is physically executable, and closure keeps all its states in Γ . If coordinate i is constant, then

$$m_i = r_i + (y_j)_i$$

along the path. Strong connectivity therefore makes y_i constant over all complexes in L . The deleted-coordinate factor

$$\prod_{i \text{ constant}} (m_i)_{y_i}$$

is a positive constant and is absorbed into the labelled rate. The remaining factor is exactly the propensity of the projected source.

An inactive linkage contributes no transition on Γ . Two projected strong linkages sharing a vertex have strongly connected union; retaining their labels and summing parallel propensities preserves the generator. The coordinate-deletion map is one-to-one on Γ , and the preceding identity intertwines the generators. Deletion and merging can only reduce the two stated counts. $\square$

Lemma 3.2 (Binary nonexplosion). *A finite binary stochastic mass-action network is nonexplosive.*

Proof. Put $N(x) = 1 + |x|_1$. A degree-two source cannot increase total population, because its binary target has degree at most two. Every population-increasing reaction therefore has source degree zero or one. Since jumps are bounded, for some $C < \infty$,

$$\sum_{y \rightarrow z} \lambda_{y \rightarrow z}(x) (|z| - |y|)^+ \leq CN(x). \quad (3)$$

Stop on first reaching $N = R$. Dynkin's formula and Gronwall give $\mathbb{E}N(X_{t \wedge \tau_R}) \leq N(x)e^{Ct}$, hence $\mathbb{P}(\tau_R \leq t) \leq N(x)e^{Ct}/R$. This tends to zero. Within a fixed population sublevel the state space is finite and the total rate is bounded, so neutral quadratic reactions cannot accumulate infinitely many jumps in finite time. $\square$

Theorem 3.3 (Intrinsic form). *Fix a closed irreducible class of a finite weakly reversible binary network. If its exact reduction in Lemma 3.1 has at most three dynamic species and at most two active linkage classes, then the original chain is positive recurrent on that class.*

The remainder of the paper proves this theorem for one and two active linkages.

4 Tier and stopped-Foster interfaces

For a population sequence x_n , deterministic tiers compare monomials $(x_n \vee 1)^y$, while stochastic tiers compare falling factorials $(x_n)_y$ and record whether y is enabled. Passing to a subsequence fixes both weak orders and the eventual coordinate caps $0, 1, \geq 2$.

Let E be the global top stochastic source tier and let a_E be the deterministic-tier level occupied by E . For a linkage support L , define

$$U_L = \{y \in L : \text{the deterministic tier of } y \text{ is at least } a_E\}. \quad (4)$$

Proposition 4.1 (Exact strong-orientation cut). *Every strongly connected orientation of the linkage supports has a deterministic-tier descending reaction sourced in E if and only if, for some linkage L ,*

$$\emptyset \neq U_L \subsetneq L, \quad U_L \subseteq E. \quad (5)$$

Proof. If (5) holds, take a directed path from U_L to its complement. The first exit is sourced in E and descends. Conversely, if (5) fails, choose in each linkage a directed Hamiltonian cycle whose unique exit from a nontrivial U_L is sourced in $U_L \setminus E$. When U_L is empty or whole there is no descending edge. The resulting orientation is strong and has no descending edge sourced in E . This is a graph construction, not an orientation enumeration. $\square$

The use of the E -level superlevel is essential. The older test using the absolute top deterministic tier is sufficient but not necessary when that tier is wholly disabled.

The class-local form of Anderson–Kim’s theorem is also needed.

Proposition 4.2 (Class-local tier Foster). *Let Γ be closed. If every proper tier sequence contained in Γ has a top stochastic source of a deterministic-tier descending reaction, then the chain is positive recurrent on Γ .*

Proof. Repeat the contradiction proof of [4, Theorem 9] with the generator restricted to Γ . Failure of a uniform entropy drift outside a finite subset gives $x_n \in \Gamma$, $|x_n| \rightarrow \infty$, and $\mathcal{AV}(x_n) > -1$. A tier subsequence remains in Γ , and the stated hypothesis forces $\mathcal{AV}(x_n) \rightarrow -\infty$, a contradiction. The published theorem is global; this paragraph is the required class-local restriction. $\square$

For stopped regions we use the following standard state-selected form.

Proposition 4.3 (Random-time Foster lemma). *Let a nonexplosive countable-state chain have a proper $W \geq 0$. Suppose that outside a finite set K , each state x is assigned an almost surely finite physical stopping time $\tau_x > 0$ such that, for fixed $\eta, \delta > 0$,*

$$\mathbb{E}_x[W(X_{\tau_x}) - W(x) + \eta\tau_x] \leq -\delta, \quad \mathbb{E}_x[W(X_{\tau_x}) + \tau_x] < \infty. \quad (6)$$

Then $\mathbb{E}_x T_K < \infty$. On a closed irreducible class, the chain is positive recurrent.

Proof. Iterate the selected rules at their actual physical endpoints and stop upon the first hit of K . Conditional expectation and localization give

$$\delta \mathbb{E}(m \wedge N_K) + \eta \mathbb{E} S_{m \wedge N_K} \leq W(x) + \delta.$$

Monotone convergence yields finite mean hitting time. From the finite target, take one ordinary jump and apply the hitting bound to its finitely many successors. A finite trace state has finite mean positive return; irreducibility then gives positive recurrence. $\square$

5 One active linkage

Theorem 5.1 (One-linkage theorem). *Let the exact reduced network on a closed irreducible class be binary, weakly reversible, have at most three dynamic species, and have exactly one active linkage. Then its chain is nonexplosive and positive recurrent for every positive rate vector.*

We summarize the proof, whose local theorems appear in Part I of the Technical Proof Supplement. On a proper escaping tier sequence, the top deterministic block is proper. If it contains an enabled source, a simple directed path to the first exit yields an all-clock access episode. For

$$G_\ell(x) = K_\ell + \sum_i \log(x_i!) + \ell \cdot x, \quad W_\ell = G_\ell^4,$$

the successful exit supplies a diverging logarithmic gap, while every competitor is paid by the exact bound

$$r\{1 + \log(1/r)\}^q \leq C_q.$$

If every top source is disabled, binarity gives a symbolic classification. With two dynamic species, the only non-deficiency-zero exception is, up to relabeling,

$$\{0, B, 2B, A + B\}.$$

With three dynamic species, the top is a mixed complex containing one zero and one divergent coordinate. There are only three nonstandard geometries: a separated singleton top, a balanced tied top, and a bounded two-disabled-top face. Their full all-clock theorems prove actual-endpoint, physical-duration, and fourth-power drift estimates. No orientation list is used in this classification or in those estimates.

A finite menu of literal statewise rules is fixed. If no menu member had uniform negative drift outside a finite set, an escaping sequence and a fixed proper-tier subsequence would contradict one of the preceding coercive local estimates. Selecting the first rule that satisfies (6) and applying Proposition 4.3 proves Theorem 5.1. The complete composition and all local calculations are Supplement Sections 1–6.

6 Two active linkages

After exact reduction and merging, the two supports are disjoint subsets of

$$\mathcal{C}_2 = \{0, A, B, C, 2A, 2B, 2C, A + B, A + C, B + C\}. \quad (7)$$

Assigning each of the ten binary complexes to linkage one, linkage two, or neither, while requiring two disjoint supports of size at least two, gives the exact ordered universe

$$3^{10} - 2\{2^{10} + 10 \cdot 2^9\} + \{1 + 10 + 10 + 90\} = 46,872. \quad (8)$$

The complete support certificate proves the pairwise-disjoint identity

$$46,872 = 27,462 + 432 + 146 + 336 + 18,496. \quad (9)$$

The five sets and their analytic theorems are summarized in Table 1. Every row is a standalone fixed-pair theorem for arbitrary strongly connected labelled orientations and arbitrary positive rates.

Table 1: Disjoint analytic partition of all 46,872 ordered support pairs.

Analytic branch	Pairs	Principal mechanism
Completed inherited mixed orbit	27,462	Valid seed theorems, then symmetry
Exclusive active-invariant orbit gap	432	Population-factorial Foster
Strictly positive invariant	146	Finite class
Level-set residual	336	Physical-time workload and service
Outside-mixed remainder	18,496	Population Foster or common marked Foster
Total	46,872	

The first row begins with the inherited 5,169 shielded/available seed pairs. The 110 seeds routed only by an invariant positive on two chart-active coordinates are removed *before* orbit closure. The remaining 5,059 seeds split exactly into 187 strictly-positive-invariant, 974 deficiency-zero, nine literal physical-seam, 1,378 corrected-tier, and 2,511 audited residual pairs. Species permutations and linkage reversal carry their completed pair theorems to exactly 27,462 pairs. The 432 symmetry images not represented by an eligible seed form the second row and have a separate global population-Foster proof. Thus no active-only invariant is used to exclude an all-three-active escape.

The 146 pairs in the third row have a common invariant with all three coefficients strictly positive, so every nonnegative integer class is finite. The 336 level-set pairs split into 312 homogeneous and 24 anisotropic supports. Their proofs use a workload-only physical-time Foster lemma, exact boundary activation kernels, and direct-death occupation; no chart-local potential is glued through an unweighted exit.

The final 18,496 pairs split disjointly into 11,842 pairs with no affine-feasible corrected-cut failure and 6,654 pairs with at least one. On the first set, a fourth power of the population factorial entropy has statewise generator drift outside a finite classwise set. On the second, one actual-target marked factorial handles every dimension: the all-active corrected cut is unconditional, all feasible two-active rows have an unconditional available/available Bellman path, and a symbolic globally-nonmixed exhaustion covers every one-active row. Its only countable Flat0 phase is closed by a cap-free killed factorial resolvent. The resulting finite menu gives one random-time Foster inequality with no chart exits or SCC quotient.

The executable certificate establishes only the finite support, tier, affine, and disjoint-union identities. Every stochastic implication in the preceding paragraphs is a theorem in the byte-pinned Technical Proof Supplement.

Theorem 6.1 (Two-linkage theorem). *Let the exact reduced network on a closed irreducible class be binary, weakly reversible, have at most three dynamic species, and have exactly two active linkages. Then its chain is nonexplosive and positive recurrent for every positive rate vector.*

Proof. Identity (9) assigns the fixed support pair to exactly one row of Table 1. Apply that row’s standalone theorem, which gives finite mean return on the fixed physical class. A support pair never moves to another row, so no potential comparison across rows is required. Lemma 3.2 supplies nonexplosion. $\square$

7 Proof of the main theorem

Proof of Theorems 1.1 and 3.3. Fix a closed irreducible class Γ and apply Lemma 3.1. Let ℓ_Γ be the number of active projected linkages. If $\ell_\Gamma = 0$, the irreducible class is a singleton. If $\ell_\Gamma = 1$, apply Theorem 5.1. If $\ell_\Gamma = 2$, apply Theorem 6.1. These cases exhaust the hypothesis. Lemma 3.2 applies throughout, and the exact conjugacy transports positive recurrence to the original physical class. $\square$

8 Reproducibility and audit boundary

The source package contains:

- (i) the main article and the complete Technical Proof Supplement;
- (ii) a SHA-256 manifest for every local proof rendered into the supplement;

- (iii) independent hostile proof audits, each pinned to exact theorem bytes;
- (iv) a corrected finite certificate for the support, tier, affine, and disjoint-union identities; and
- (v) focused tests that reproduce the finite counts and fingerprints.

The executable layer has a deliberately narrow claim boundary. It neither samples nor enumerates arbitrary reaction orientations, stochastic histories, or population states. All rate, orientation, endpoint, and physical-time quantifiers are discharged in the mathematical proofs.

The older 2,511-based global composition is retained only as provenance. It correctly certifies its literal seed-level residual but does not exhaust the full support universe after species and linkage symmetry. The publication dependency is now the exact 46,872-pair identity (9) and the five standalone pair theorems in Table 1.

References

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